

Insight Summary Report: Apple Inc. Stock Market Analysis Using Python

Dataset Overview

This project analyzed Apple Inc. historical stock market data over a 5-year period using Python. The dataset was obtained from Yahoo Finance and contains daily stock market records for Apple Inc. The main features in the dataset include Date, Open Price, High Price, Low Price, Close Price, Adjusted Close Price, and Trading Volume.

The dataset contained 1,255 rows and 7 original columns. Each row represents one trading day. Since stock markets do not open every day, weekends and market holidays are not included in the dataset.

Data Cleaning and Transformation

The dataset was checked for missing values, duplicate records, and incorrect data types. No missing values or duplicate rows were found. The Date column was already in datetime format, while the price columns were stored as numerical values and the Volume column was stored as an integer.

The data was sorted by Date to ensure that it followed the correct time-series order from oldest to newest. This

was important because time-series analysis depends on the correct arrangement of dates.

Several new columns were created to support deeper analysis. These included Year, Month, Month Name, Daily Price Change, Daily Percentage Change, 7-Day Moving Average, 30-Day Moving Average, Trading Range, and 30-Day Rolling Volatility.

Time-Series and Feature Engineering Techniques Applied

The analysis involved several advanced Pandas and time-series techniques. Date-based features were extracted from the Date column to support monthly and yearly analysis. Daily Price Change was created by subtracting the opening price from the closing price, while Daily Percentage Change was calculated to measure the stock's daily return compared to the previous trading day.

Moving averages were created using rolling calculations. The 7-day moving average was used to observe short-term stock movement, while the 30-day moving average helped reveal the broader stock price trend. A 30-day rolling volatility feature was also created to measure periods of higher and lower price fluctuation.

Monthly analysis was also performed by grouping the data by year and month. This helped summarize Apple's

performance at a broader level and identify strong and weak months.

Trends Identified

The closing price trend showed that Apple's stock moved upward overall over the 5 years. Although the stock experienced some declines and fluctuations, especially around late 2022 to early 2023 and around 2025, it recovered strongly and reached higher price levels by 2026.

The moving average analysis confirmed this overall upward trend. The 7-day moving average closely tracked the daily closing price, while the 30-day moving average provided a smoother view of the broader trend. From 2023 onward, the moving averages showed stronger upward movement.

The trading volume analysis showed that Apple's stock experienced several noticeable spikes in trading activity. These spikes suggest periods when investor activity increased, possibly due to market events, earnings announcements, or broader market reactions.

Key Insights

Apple's highest average closing price occurred in July 2026, with an average close of about \$310.18. However,

July 2026 may represent partial-month data, so this result should be interpreted carefully.

The lowest average closing price occurred in January 2023, with an average close of about \$135.78. This suggests that early 2023 was one of the weaker periods in the 5-year dataset.

December 2021 recorded the highest trading volume, with total volume of about 2.44 billion shares. This indicates a period of strong market activity around Apple's stock.

Monthly return analysis showed that July 2022 had the strongest monthly performance, with a return of about 18.86%. On the other hand, December 2022 had the weakest monthly return, with a decline of about -12.23%.

The volatility analysis showed that Apple's stock experienced periods of higher price fluctuation, especially around 2025. This means that even though the overall long-term trend was upward, the stock still experienced periods of increased uncertainty and stronger price movement.

Recommendations

Investors and analysts should not rely only on daily closing prices when studying stock performance. Moving averages should also be monitored because they help reveal clearer short-term and medium-term trends.

Monthly returns should be reviewed to identify periods of strong and weak stock performance. This gives a better understanding of how the stock performs over time instead of focusing only on daily changes.

Volatility should also be considered when analyzing risk. Even when a stock shows long-term growth, periods of high volatility may indicate increased uncertainty or stronger market reactions.

Trading volume should be analyzed together with price movement because volume spikes can show periods of increased investor interest or market activity.

For deeper future analysis, Apple's stock performance can be compared with other technology companies or broader market indexes such as the NASDAQ or S&P 500.

Conclusion

This project applied advanced Python techniques to analyze Apple Inc. historical stock market data. The analysis included data cleaning, transformation, feature engineering, time-series analysis, and visualization.

The project showed that Apple's stock had an overall upward trend over the 5 years, although it experienced periods of decline, volatility, and fluctuating monthly returns. By creating features such as moving averages, monthly returns, trading range, and rolling volatility, the

analysis provided a deeper understanding of Apple's stock behavior.

Overall, this project demonstrated how Python and Pandas can be used to transform raw stock market data into meaningful insights for financial and time-series analysis.